Sampling Theorem

Course: Math 414

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Core Idea

Human ears can only hear things under 20kHz. We have no way to record continuous audio, so what do we do?

We can sample a signal at a frequency greater than twice its highest frequency component and get an accurate sample.

Definition

f is frequency band limited if there is a constant $\Omega > 0$ s.t.

$$\hat{f}(\lambda) = 0 \quad |\lambda| > \Omega \tag{1}$$

Some variables to know: v is the frequency in Hz

To get from standard frequency v to angular frequency Ω,

$$\Omega = 2\pi v \tag{2}$$

Nyquist Frequency: $v = \frac{\Omega}{2\pi}$

Nyquist Rate (the minimum required sampling speed): $2v = \frac{\Omega}{\pi}$

This is important the Nyquist rate is the minimum speed at which you must take samples of an analog signal

This just basically means real time. It's like a clock with two hands

Shannon-Whittaker Sampling Theorem

$\hat{f}(\lambda)$ is piecewise smooth, continuous, and band limited

$f = \mathcal{F}^{-1}\left[\hat{f}\right]$ is completely determined by its values at the points $t_j = j\frac{\pi}{\Omega}; j \in \mathbb{Z}$

In math, it means

$$f(t) = \sum_{j=-\infty}^{\infty} f\left(\frac{j\pi}{\Omega}\right) \frac{\sin(\Omega t - j\pi)}{\Omega t - j\pi} \tag{3}$$

If you notice, we need to sample each $\frac{\pi}{\Omega}$ seconds, which is our period, so the reverse is the Nyquist rate

Proof idea (Need to know for exam)

We know that $\hat{f}$ is continuous on $[-\Omega, \Omega]$ by the definition of band limited. Let's expand $\hat{f}$ as a Fourier series on $[-\Omega, \Omega]$ (since it's 0 outside of this anyways)

$$\hat{f}(\lambda) = \sum_{k=-\infty}^{\infty} c_k e^{(i\pi k\lambda)\Omega} \tag{4}$$

$$c_k = \frac{1}{2\Omega} \int_{-\Omega}^{\Omega} \hat{f}(\lambda) e^{-\frac{i\pi k\lambda}{\Omega}} \, \mathrm{d}\lambda \tag{5}$$

We want to change this to look like a Fourier Transform. Since it's band limited, we can say that our integral goes everywhere (since it's 0 everywhere other than our band, so it's okay)

$$c_k = \frac{\sqrt{2\pi}}{2\Omega} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\lambda) e^{-\frac{i\pi k\lambda}{\Omega}} \, \mathrm{d}\lambda \tag{6}$$

$$c_k = \frac{\sqrt{2\pi}}{2\Omega} f\left(-\frac{k\pi}{\Omega}\right) \tag{7}$$

Let's plug this back in

$$\hat{f}(\lambda) = \sum_{j=-\infty}^{\infty} \frac{\sqrt{2\pi}}{2\Omega} f\left(-\frac{j\pi}{\Omega}\right) e^{\frac{i\pi j\lambda}{\Omega}} \tag{8}$$

This converges uniformly since we said $\hat{f}$ is piecewise smooth, continuous and it converges uniformly.

Let's also change the summation index from j to $j = -k$

$$\hat{f}(\lambda) = \sum_{j=-\infty}^{\infty} \frac{\sqrt{2\pi}}{2\Omega} f\left(\frac{j\pi}{\Omega}\right) e^{-\frac{i\pi j\lambda}{\Omega}} \tag{9}$$

If we now take the I.F.T. of $f(t)$, and plug in what we found for $\hat{f}$, we end up with the following:

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\Omega}^{\Omega} \hat{f}(\lambda) e^{i\lambda t} \, \mathrm{d}\lambda \tag{10}$$

If we plug in what we got for $\hat{f}$

$$f(t) = \frac{1}{\cancel{\sqrt{2\pi}}} \int_{-\Omega}^{\Omega} \left(\sum_{j=-\infty}^{\infty} \frac{\cancel{\sqrt{2\pi}}}{2\Omega} f\left(\frac{j\pi}{\Omega}\right) e^{-\frac{i\pi j\lambda}{\Omega}} \right) e^{i\lambda t} \, \mathrm{d}\lambda \tag{11}$$

Swapping around the sum and integral

$$f(t) = \sum_{j=-\infty}^{\infty} \frac{1}{2\Omega} f\left(\frac{j\pi}{\Omega}\right) \int_{-\Omega}^{\Omega} e^{i\lambda\left(t-\frac{\pi j}{\Omega}\right)} \, \mathrm{d}\lambda \tag{12}$$

The integral simplifies to

$$\int_{-\Omega}^{\Omega} e^{i\lambda\left(t-\frac{\pi j}{\Omega}\right)} \, \mathrm{d}\lambda = 2\Omega \frac{\sin(t\Omega - j\pi)}{t\Omega - j\pi} \tag{13}$$

If we plug things back in

$$f(t) = \sum_{j=-\infty}^{\infty} \cancel{\frac{1}{2\Omega}} f\left(\frac{j\pi}{\Omega}\right) \left(\cancel{2\Omega} \frac{\sin(t\Omega - j\pi)}{t\Omega - j\pi}\right) \tag{14}$$

This brings us back to the Shannon thing.

What happens if you over/under sample?

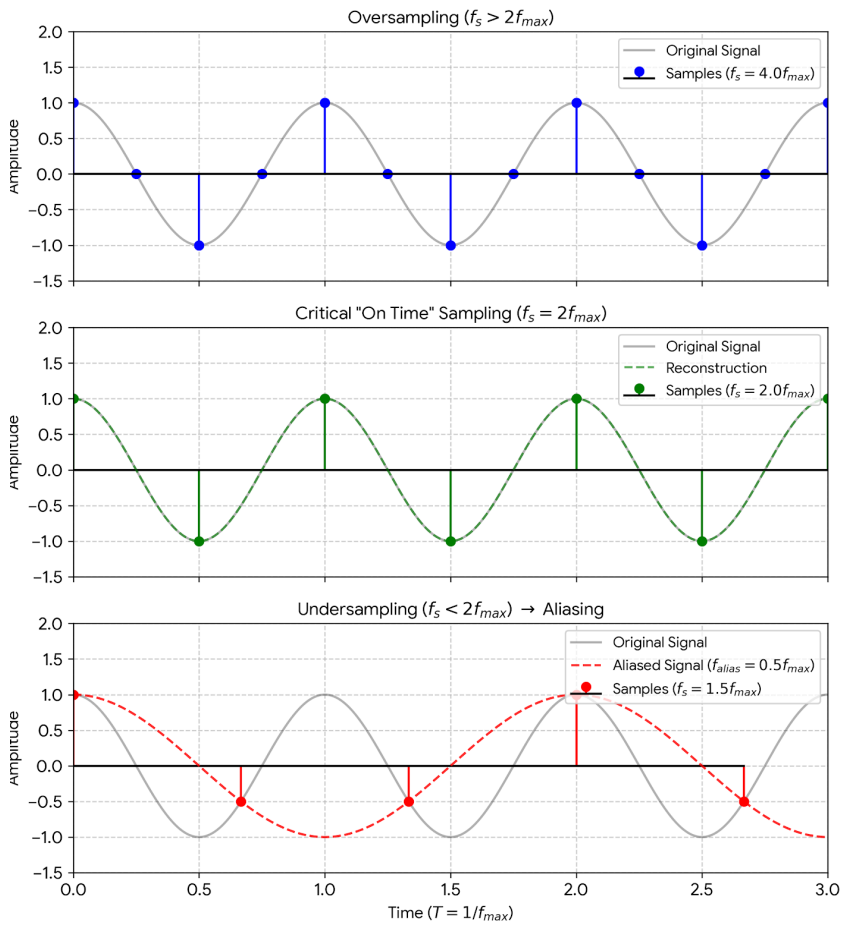

Figure 1: We can see the effects of over/under sampling in reconstructing our original signal here